CHERI FAVEL



Chemical Engineer (MEng) | Process Development & Industrial R&D

Abu Dhabi, UAE | UAE resident | Available immediately | Open to relocation

cherifavel@gmail.com | +971 58 584 5473 | linkedin.com/in/cheri-favel | Engineering portfolio

PROFESSIONAL PROFILE

MEng-qualified Chemical Engineer with experience in industrial research and development, process engineering and applied chemical process development. Skilled in designing and executing experimental programmes, analysing process performance, conducting engineering calculations and translating technical findings into practical recommendations. Experience spans hydrometallurgy, electrochemical systems, separation processes, resource recovery and industrial water treatment. Published researcher with strong analytical, technical reporting and problem-solving capabilities. Interested in process engineering, process development and industrial R&D opportunities across chemical manufacturing, energy and related industries.

TECHNICAL CAPABILITIES

- **Process engineering:** mass and energy balances, process calculations, PFDs/P&IDs, equipment-sizing support, design-basis review, specifications and supplier quotation review.
- **Process development:** experimental design, leaching, selective precipitation, electrowinning and electrolysis, laboratory reactor testing and techno-economic evaluation support.
- **Laboratory and data:** hands-on ICP-OES, particle-size and pH analysis; interpretation of XRF, XRD, ATR-FTIR, SEM-EDS, BET and TGA results; kinetic modelling and technical reporting.
- **Software:** Excel, VBA, Power Query, PivotTables, Visio and Python; foundational Aspen Plus/HYSYS exposure.

ENGINEERING AND RESEARCH EXPERIENCE

Process & R&D Engineer
Mar 2021 – Jul 2022

Free Radical Process Design (FRPD), South Africa

- Conducted laboratory and engineering investigations for hydrometallurgical, electrochemical and industrial water-treatment process development.
- Prepared process calculations, mass and energy balances, flow diagrams (PFDs/P&IDs) and technical reports; reviewed and revised design bases; supported equipment sizing and supplier quotation review.
- **Manganese processing:** contributed to design-basis and process documentation for a manganese sulphate monohydrate demonstration plant; ran selective-precipitation investigations.
- **Electrowinning:** ran copper and cobalt trials on a continuous rotating-cathode unit, investigating deposit formation and release; contributed to a proposed zinc-recovery application.
- **Zinc electrolysis:** planned, set up, ran and reported plate-electrode zinc electrolysis trials end to end, including experimental design, sampling, analysis and recommendations.
- **Gold recovery:** ran laboratory electrowinning/sludging reactor trials on refinery eluate, varying current density, voltage, flow rate and residence time; supported mass-balance and ICP-OES/XRF data analysis.
- **Rare earths:** ran lanthanide precipitation experiments and contributed calculations to a CAPEX/OPEX model for a neodymium metal plant.
- **Industrial water:** investigated metal removal from synthetic mine-tailings water using oxidation and selective precipitation; wrote the laboratory-trial section of a research proposal for Fe, Zn and Mn recovery from Peruvian tailings water.

Postgraduate Researcher in Chemical Engineering
Jan 2020 – Dec 2021

University of Pretoria, South Africa

- Designed nitric-acid leaching experiments on phlogopite and characterised mineral feed and residues; identified diffusion-controlled kinetics (one-dimensional D1 model) and determined activation energies; research published in *Clays and Clay Minerals* (2022).
- Planned laboratory work, maintained experimental records, guided undergraduate laboratory work and communicated results through scientific reports.

ADDITIONAL PROFESSIONAL EXPERIENCE

Business Setup & Marketing Consultant Apr 2025 – Present
Cresco Consulting FZCO, UAE

- Prepare and submit company-registration documentation through the IFZA portal; manage client information and records; maintain the company website and creative content.

Engineer Assistant / Deckhand / Stewardess May 2023 – Apr 2025
Research Vessel Rosa, United States & Bahamas

- Assisted with pumps, plumbing and RO watermaker maintenance and troubleshooting; organised technical drawings, maintenance records and procurement within marine safety procedures.

Academic Journal Editor Nov 2022 – Apr 2023
CACTUS Communications, remote

- Edited scientific manuscripts for technical clarity, consistency and journal requirements; reviewed terminology and provided structured author feedback.

Teaching Assistant (Part-Time) Feb 2019 – Jun 2021
University of Pretoria, South Africa

- Supported laboratory sessions and tutoring in particle technology, kinetics and engineering practice/costing; supervised practical work and assessed student submissions.

Online English Language Consultant Sep 2020 – Mar 2021
MPC Connect, South Africa / remote

- Delivered structured lessons and feedback to non-native English speakers, developing clear communication and independent remote-working skills.

Laboratory Intern (Vacation Work) Dec 2017 – Jan 2019
University of Pretoria, South Africa

- Completed 12 weeks of laboratory placements in mineral leaching and fertiliser production, including experimental optimisation and analytical testing.

EDUCATION AND RESEARCH

MEng Chemical Engineering | University of Pretoria | 2021

BEngHons Chemical Engineering with distinction | University of Pretoria | 2020

BEng Chemical Engineering | University of Pretoria | 2019

- **MEng dissertation:** Development of Kinetic Parameters for Nitric Acid Leaching of Phlogopite and Characterization of the Solid Residue.
- **Publication:** Favel, C.M. & du Plessis, B.J. (2022). Development of Kinetic Parameters for Nitric Acid Leaching of Phlogopite and the Characterization of Solid Products. *Clays and Clay Minerals*, 70, 106–119. doi.org/10.1007/s42860-022-00180-x
- **Environmental Engineering specialisation:** water and effluent treatment, bioreactors, air pollution, waste remediation and environmental regulation.

ResearchGate: researchgate.net/profile/Cheri-Favel

RELEVANT PROFESSIONAL DEVELOPMENT

- **SAMA Return-to-Work Programme, Phase 2** | Women's Pavilion, Expo City Dubai (with HSBC) | Nov 2026
- **LEED Green Associate Exam Preparation** | LinkedIn Learning | Sep 2026 (exam preparation course)
- **Project Management: Energy and Water Projects** | PMI / LinkedIn Learning | Feb 2026
- **Occupational Safety and Health: Hazard Communication** | LinkedIn Learning | Feb 2026
- **Python Mega Course: Developing Apps and AI Agents** | Udemy | In progress
- **Maritime safety**: STCW Basic Safety | Dec 2023 · Approved Engine Course (AEC 1) | Feb 2024

LANGUAGES

English (native) | Afrikaans (proficient) | Hebrew (basic)

SCAN OR CLICK TO CONNECT

Contact details, portfolio & research

Opens my portfolio page with project summaries, publication links and contact details.